Census stereo: method and results

Problem 1. Compare every neighbor with its center using <, omit the center, and encode comparisons in a 64-bit census descriptor. For each left pixel, test right pixel x-d, sum descriptor Hamming distances over WA x WA, then select the lowest cost (ties choose smaller d). Search d=0..63. Reject candidates whose census or aggregation windows cross either image boundary. Pixels without candidates are black.

Census display. Show descriptor distance from the all-zero descriptor (set-bit count), mapping the complete possible range 0..8 or 0..48 to 0..255. Both sizes show similar relative-brightness structure, but 7 x 7 uses more context and has finer count levels. These displays discard bit arrangement; matching uses every bit.

Evaluation. Ground truth is floor(stored/4 + 0.5). Absolute error >1 is bad. Whole image counts all 168,750 pixels, including invalid black borders and GT zeros. Shared interior uses the same 131,311 pixels: rows [7,368), columns [70,443), GT>0. GT-zero exclusion and border treatment are explicit assumptions requiring course confirmation.

Wc	WA	Whole image error	Shared interior error
3	5	27.9911%	18.8012%
3	9	24.4480%	13.1459%
7	5	24.1807%	12.8687%
7	9	24.5031%	11.6175%

Problem 2. Interpret quarter resolution as quarter total pixels: 3072 x 2048 becomes 1536 x 1024. Scale both first rows of K by 0.5: fx=1379.75, fy=1382.10, cx=760.35, cy=503.40. Image dimensions/focal lengths and normalized principal point remain constant, preserving both fields of view. This resolution interpretation still needs confirmation; scale 0.25 is configurable.

Geometry. Use R=I, left center (0,0,0), right center (0.12,0,0). World-to-camera t values are (0,0,0) and (-0.12,0,0). Shared fixed K and parallel axes give equal vertical coordinates. The baseline targets about 43 pixels at the closest surface: d=fx B/Z. Scene Z=3.8663..9.6567 predicts d=17.1456..42.8241. Search 0..44 covers depths from 3.7630 toward infinity (d=0 is the infinite-depth limit). All lengths are scene units; physical scale is unconfirmed.

Rendering and findings. NumPy pinhole projection with a nearest-depth buffer replaces unsupported Open3D headless rendering on this Mac. No maximum filter or hole filling is used. Grayscale is round(0.30R + 0.59G + 0.11B). Run the same unfiltered matcher for all four settings. Against a depth-derived reference on 1,379,716 mutually visible interior pixels, errors are 0.7700%, 0.2253%, 0.3017%, 0.2886% in (3,5), (3,9), (7,5), (7,9) order. This synthetic diagnostic excludes holes and is not independent ground truth. Larger windows reduce speckle but blur boundaries. New viewpoints expose missing surfaces; occlusions remain difficult.

Verification. Hand-computed census, direct Hamming/window sums, shifted and identical inputs, brute-force WTA, rectification, RGB readback, and calibration checks passed. Reproduce with teddy.py and fish.py using the documented Python environment. Figures follow; optional refinements are separate from these required results.

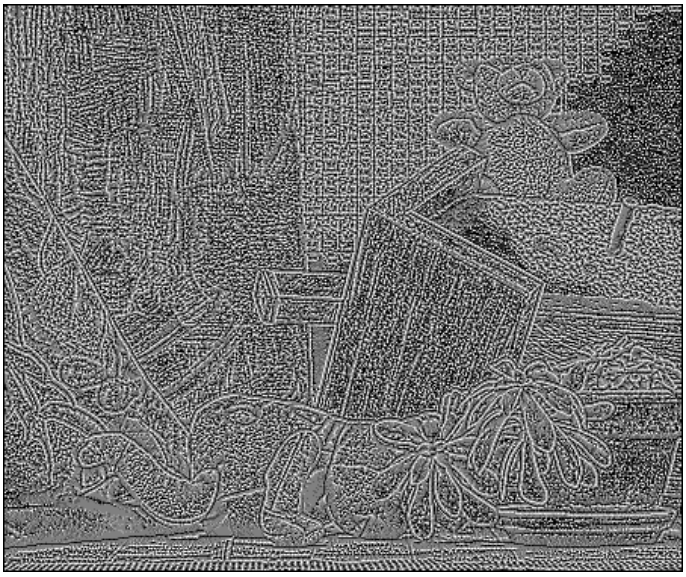
Teddy: census visualizations

Black to white: set-bit count 0..8 (Wc=3) or 0..48 (Wc=7).

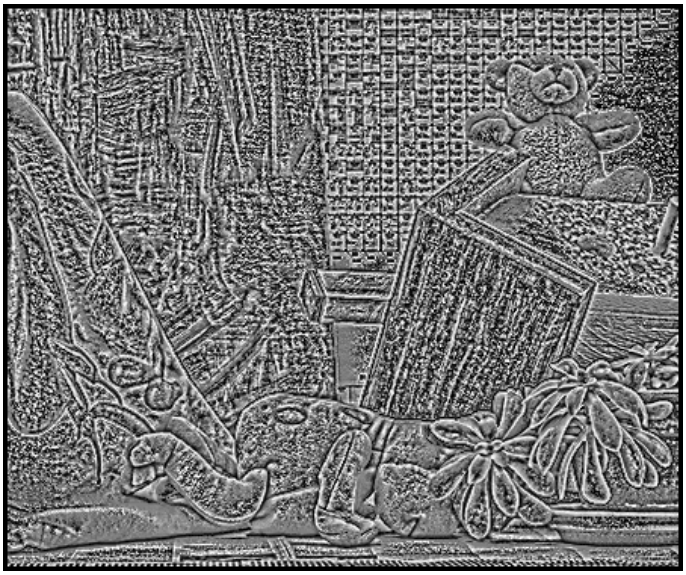
Left | Wc = 3



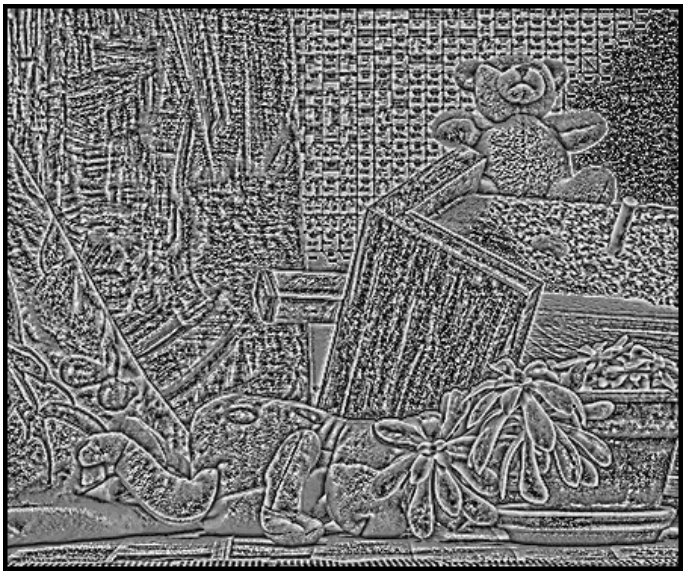
Right | Wc = 3



Left | Wc = 7



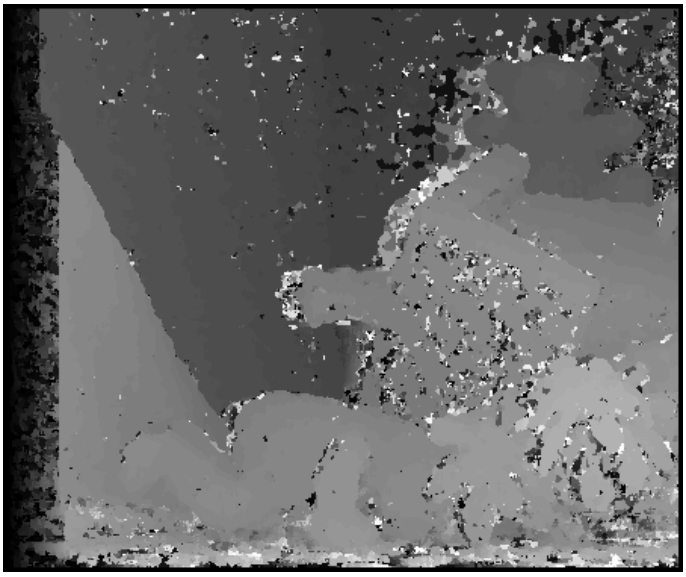
Right | Wc = 7



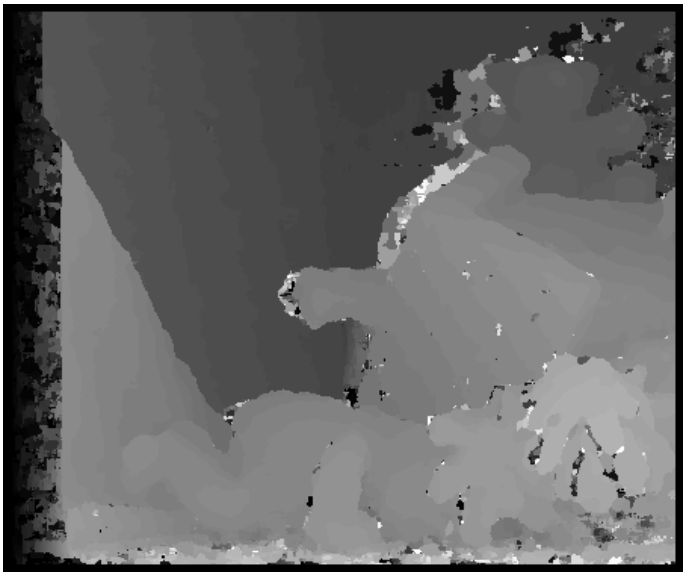
Teddy: disparity maps

Black to white: disparity 0..63 pixels. Invalid borders are also black.

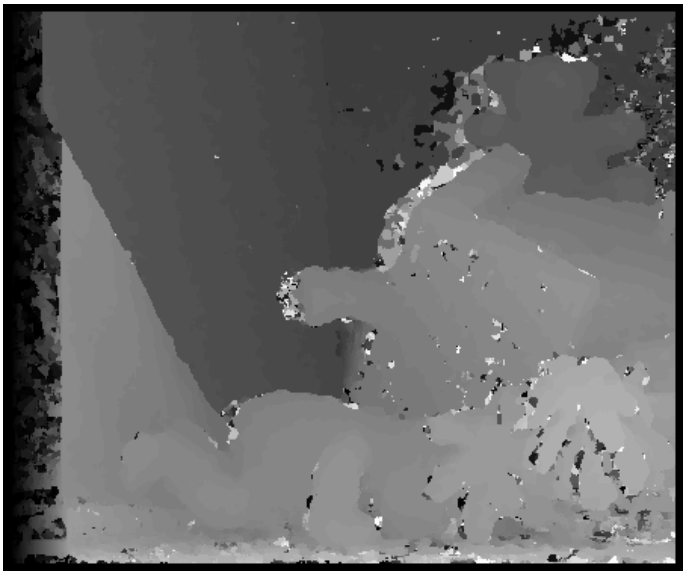
Wc = 3, WA = 5



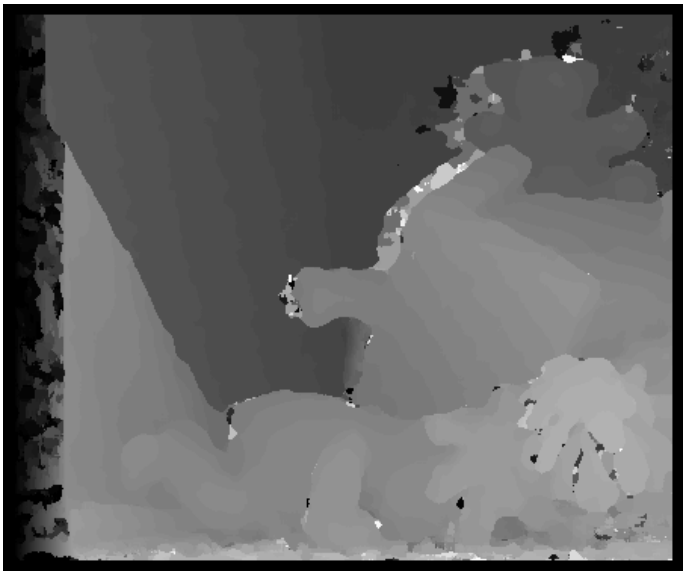
Wc = 3, WA = 9



Wc = 7, WA = 5



Wc = 7, WA = 9



Fish: rectified stereo images

1536 x 1024 pixels per view | Baseline 0.12 scene units | No hole filling.

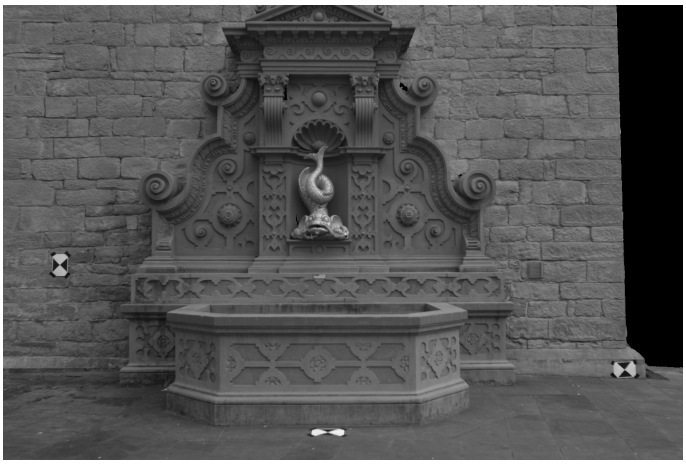
Left | RGB



Right | RGB



Left | GRAY



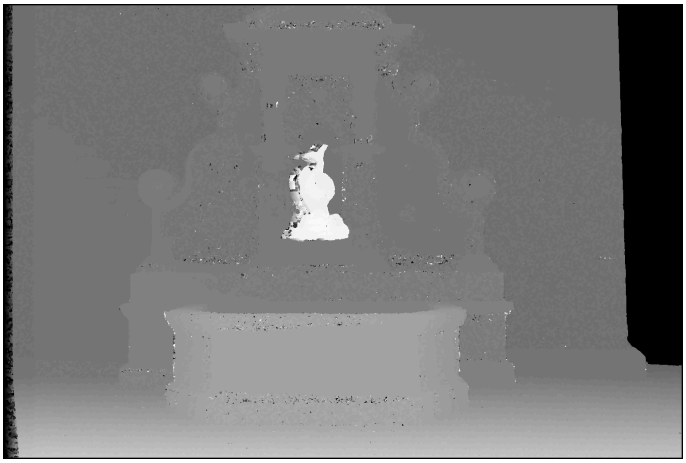
Right | GRAY



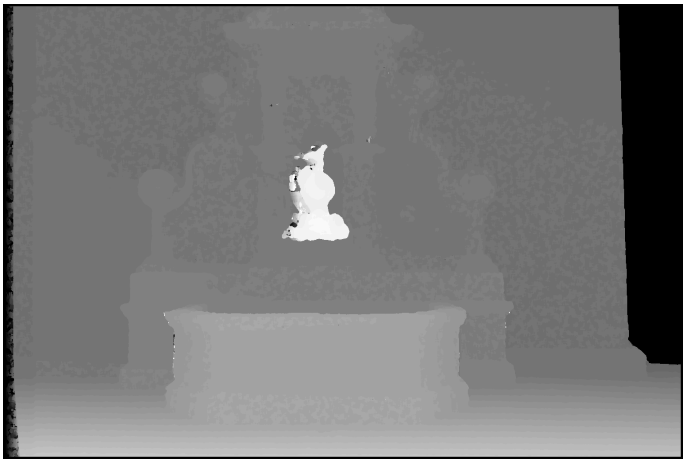
Fish: disparity maps

Black to white: disparity 0..44 pixels. Invalid borders and unsupported regions remain.

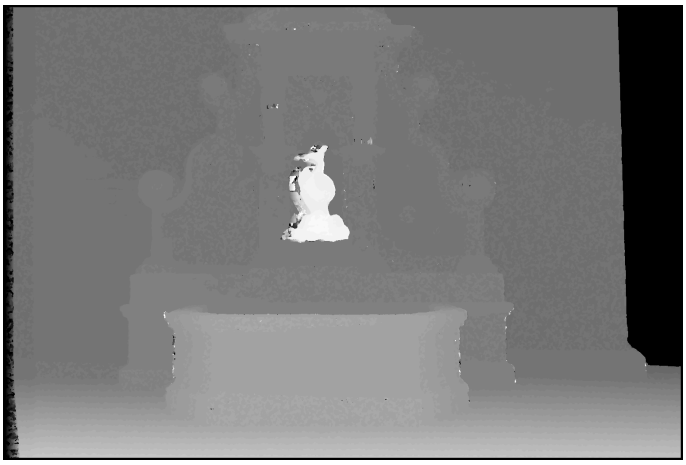
Wc = 3, WA = 5



Wc = 3, WA = 9



Wc = 7, WA = 5



Wc = 7, WA = 9

